

Lessons

- Table of Contents
- Confidence Levels
- Notes
- Bookmarks
- Highlights

Multiplying Rational Expressions

Multiplying Rational Expressions

Toolbox



Multiplying Rational Expressions

The rules for multiplying rational expressions are the same as the rules for multiplying fractions. First, look for ways to cancel. Then multiply the numerators and the denominators straight across.

Example 1

Hide

Multiply: $\frac{6k^2}{5} \cdot \frac{4m^2}{18km}$

$\frac{m^2}{m} = m$

$\frac{k^2}{k} = k$

$\frac{18}{6} = 3$

Step 1

Cancel. Remember to look for common factors in the coefficients and the variables.

$\frac{\overset{k}{\cancel{6k^2}}}{5} \cdot \frac{\overset{m}{\cancel{4m^2}}}{\underset{3}{\cancel{18km}}}$

Step 2

Multiply straight across.

$\frac{\overset{k}{\cancel{6k^2}}}{5} \cdot \frac{\overset{m}{\cancel{4m^2}}}{\underset{3}{\cancel{18km}}} = \frac{\boxed{4km}}{\boxed{15}}$

You can also cancel binomial factors as long as the factors are identical. After you finish multiplying, use FOIL or the distributive property to simplify.

Example 2

Hide

Multiply: $\frac{(x+3)(\cancel{x-2})}{(x+1)} \cdot \frac{(x+5)}{(\cancel{x-2})}$

Step 1

Cancel.

$\frac{(x+3)(\cancel{x-2})}{(x+1)} \cdot \frac{(x+5)}{(\cancel{x-2})}$

Step 2

Multiply straight across and simplify.

$\frac{(x+3)(\cancel{x-2})}{(x+1)} \cdot \frac{(x+5)}{(\cancel{x-2})} = \frac{(x+3)(x+5)}{(x+1)} = \frac{x^2+8x+15}{x+1}$

Restrictions:

$x+1 \neq 0$

$x \neq -1$

$x-2 \neq 0$

$x \neq 2$

$x^2+5x+3x+15$

The numerators and denominators must be fully factored before you can cancel.

Example 3

Hide

Multiply: $\frac{x^2-12x+35}{x^2+2x} \cdot \frac{x^2-4}{x^2-6x-7}$

Step 1

Factor.

$\frac{x^2-12x+35}{x^2+2x} \cdot \frac{x^2-4}{x^2-6x-7} = \frac{(x-7)(x-5)}{x(x+2)} \cdot \frac{(x-2)(x+2)}{(x-7)(x+1)}$

Step 2

Cancel and then multiply. Simplify as needed. Leave the denominator factored.

$\frac{(\cancel{x-7})(x-5)}{x(\cancel{x+2})} \cdot \frac{(x-2)(\cancel{x+2})}{(\cancel{x-7})(x+1)} = \frac{(x-5)(x-2)}{x(x+1)} = \frac{x^2-7x+10}{x(x+1)}$

Restrictions:

$x \neq 0$

$x+1 \neq 0$

$x \neq -1$